and use our logic to understand the exact meaning of the text. For example, scholars who wished to know the shape of the earth scanned the Bible looking for relevant references. One pointed out that in Job 38:13, it says that God can 'take hold of the edges of the earth, and the wicked be shaken out of it'. This implies – reasoned the pundit – that because the earth has 'edges' of which we can 'take hold', it must be a flat square. Another sage rejected this interpretation, calling attention to Isaiah 40:22, where it says that God 'sits enthroned above the circle of the earth'. Isn't that proof that the earth is round? In practice, that meant that scholars sought knowledge by spending years in schools and libraries, reading more and more texts, and sharpening their logic so they could understand the texts correctly.

The Scientific Revolution proposed a very different formula for knowledge: **Knowledge = Empirical Data × Mathematics**. If we want to know the answer to some question, we need to gather relevant empirical data, and then use mathematical tools to analyse the data. For example, in order to gauge the true shape of the earth, we can observe the sun, the moon and the planets from various locations across the world. Once we have amassed enough observations, we can use trigonometry to deduce not only the shape of the earth, but also the structure of the entire solar system. In practice, that means that scientists seek knowledge by spending years in observatories, laboratories and research expeditions, gathering more and more empirical data, and sharpening their mathematical tools so they could interpret the data correctly.

The scientific formula for knowledge led to astounding breakthroughs in astronomy, physics, medicine and countless other disciplines. But it had one huge drawback: it could not deal with questions of value and meaning. Medieval pundits could determine with absolute certainty that it is wrong to murder and steal, and that the purpose of human life is to do God's bidding, because scriptures said so. Scientists could not come up with such ethical judgements. No amount of data and no mathematical wizardry can prove that it is wrong to murder. Yet human societies cannot survive without such value judgements.